

MAT 310, Handout 26: Order Statistics (6.7)

Motivation: Given a collection of RVs, we want to know the distribution of the maximum and minimum of the RVs.

Example 1. If Y_1 and Y_2 are independent $U(0, 1)$ RVs, find the density function of $X = \max\{Y_1, Y_2\}$.

Order Statistics: Let $Y_1, Y_2, \dots, Y_n$ be independent continuous random variables with cdf $F(y)$ and pdf $f(y)$. The *order statistics* are given by:

$$Y_{(1)} \leq Y_{(2)} \leq \dots \leq Y_{(n)},$$

where $Y_{(1)} = \min\{Y_1, Y_2, \dots, Y_n\}$ and $Y_{(n)} = \max\{Y_1, Y_2, \dots, Y_n\}$. Therefore,

$$\begin{aligned} F_{Y_{(n)}}(y) &= P(Y_{(n)} \leq y) \\ &= \\ &= \\ &= \end{aligned}$$

And so,

$$f_{Y_{(n)}}(y) =$$

Also,

$$\begin{aligned} F_{Y_{(1)}}(y) &= P(Y_{(1)} \leq y) \\ &= \\ &= \\ &= \\ &= \end{aligned}$$

And so,

$$f_{Y_{(1)}}(y) =$$

Example 2. If Y_1 , Y_2 , and Y_3 are independent $\exp(\beta)$ RVs, find:

(a) the density of $W = \max\{Y_1, Y_2, Y_3\}$

(b) the density of $X = \min\{Y_1, Y_2, Y_3\}$